UNIT - IV

Strategic & Project Management

Syllabus at a glance

- Vision, mission, goals - Corporate planning process - Environmental scanning - Strategy formulation and implementation - SWOT
- Project management - Network analysis - PERT and CPM - Critical path - Probability of completion - Project cost analysis and crashing

1. Strategic Management - Meaning

A strategy is a long-term, comprehensive plan of action to achieve organisational objectives by matching the organisation's resources with its environment. (Greek: strategos = "art of the general".)

- Alfred Chandler : "Strategy is the determination of the basic long-term goals of an enterprise and the adoption of courses of action and allocation of resources necessary to carry out these goals."
- Glueck : a unified, comprehensive and integrated plan relating the strategic advantages of the firm to environmental challenges
- Strategic management = continuous process of formulating, implementing and evaluating strategies to achieve long-term success

Features of strategy

- Long-term and future oriented - Top-management decision - Matches resources with environment - Gives competitive advantage - Involves major resource commitment - Flexible and reviewed

Levels of strategy

Level	Who or what	Example
Corporate	Board or CEO - which businesses to be in (growth, diversify, retrench)	Tata group entering EVs
Business (SBU)	Business head - how to compete in one business	Low-cost vs premium positioning

Level	Who or what	Example
Functional	Dept. heads - marketing, finance, HR, production support	New distribution plan, hiring plan

2. Vision, Mission, Goals & Objectives



Vision

- A vision is a vivid picture of the future the organisation wants to create - inspiring, long-term, short and memorable. Example: "To be the most trusted car maker in India."

Mission

- A mission states the organisation's purpose : what business we are in, whom we serve, how we are different, our values. Example: "We make safe, affordable cars for Indian families through innovation and quality."
- Good mission : clear, realistic, customer-focused, motivating, includes products, markets, technology, philosophy, concern for employees and society

Basis	Vision	Mission
Answers	Where do we want to go?	Why do we exist or what do we do?
Time	Future (10+ years)	Present and near future
Nature	Aspirational	Practical, action-oriented

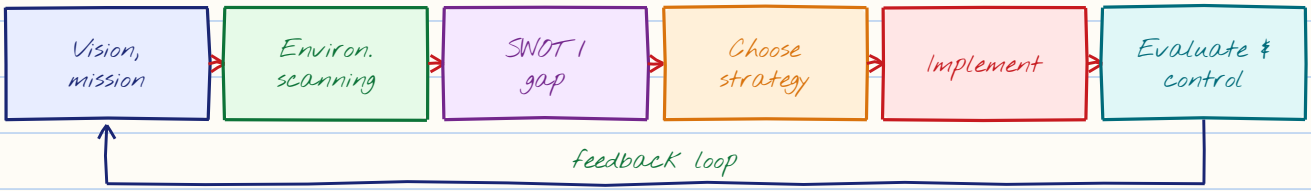
Goals & Objectives

- Goals : broad, qualitative or long-term ends (become market leader) ; Objectives : specific, quantitative and time-bound (raise sales by 15% in FY 2027-28)
- SMART objectives : Specific, Measurable, Achievable, Relevant, Time-bound

- Types : economic (profit, market share), social (CSR), human (training), national (exports) ; Drucker's 8 key result areas : market standing, innovation, productivity, resources, profitability, manager performance, worker performance, public responsibility

3. Corporate Planning Process

Corporate planning = formal, systematic process of deciding the long-range direction of the whole organisation and the means to reach it.



1. Define vision, mission, objectives
2. Environmental scanning - find opportunities and threats ; Internal appraisal - find strengths and weaknesses
3. Strategic gap analysis or SWOT - where we are vs where we want to be
4. Formulate alternative strategies and evaluate them
5. Choose the best strategy - by risk, return, fit, resources
6. Prepare medium and short-range plans, budgets, policies (derived plans)
7. Implement and monitor ; take corrective action (control) - feedback to step 1

4. Environmental Scanning

Environmental scanning = continuously monitoring, collecting and analysing information about forces outside (and inside) the organisation to spot opportunities and threats early (Aguilar).

Environment

- External - Macro (PESTLE) : Political, Economic, Social, Technological, Legal, Environmental
- External - Micro (task or industry) : customers, suppliers, competitors, substitutes, new entrants, banks, govt. regulators - analysed by Porter's 5

Forces

- Internal : resources, people, structure, culture, technology, finance, marketing, R&D → strengths and weaknesses

Techniques

- ETOP (Environmental Threat and Opportunity Profile) - SWOT - PESTLE - Porter's 5 forces - BCG matrix - scenario building - Delphi - industry and competitor analysis

Porter's 5 forces (industry attractiveness)

- Threat of new entrants - Bargaining power of buyers - Bargaining power of suppliers - Threat of substitutes - Rivalry among existing competitors

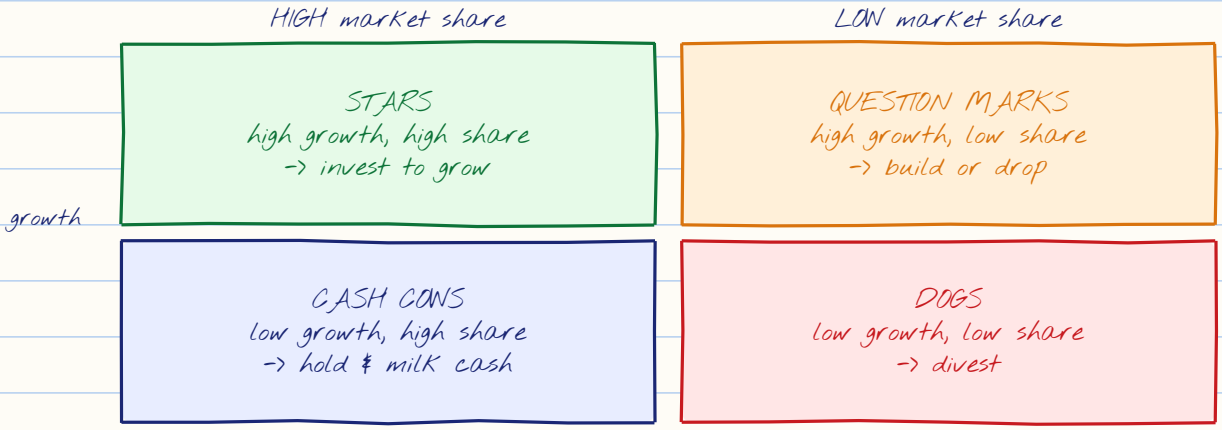
5. Steps in Strategy Formulation & Implementation

Formulation

1. Set or review mission and objectives 2. Environmental and internal analysis (SWOT) 3. Identify strategic alternatives 4. Evaluate alternatives (criteria: fit, feasibility, risk, return) 5. Select strategy 6. Prepare strategic plan

Grand (corporate) strategies - Ansoff / Glueck

- Stability - maintain present business, marginal improvements - Expansion (growth) - market penetration, market development, product development, diversification , mergers and acquisitions, joint ventures
- Retrenchment - turnaround, divestment, liquidation - Combination - mix of the above
- Porter's generic strategies : Cost leadership , Differentiation , Focus
- BCG matrix : Stars (high growth - high share), Cash cows, Question marks, Dogs → invest or hold or harvest or divest



Implementation (put the strategy into action)

- Project implementation (set up facilities, schedules) - Procedural implementation (laws, licences) - Resource allocation (money, people, time)
- Structural implementation (organisation design, delegation) - Behavioural (leadership, culture, motivation, change management) - Functional (policies for marketing, finance, HR, production)
- Control and evaluation : balanced scorecard, KPIs, budgets, audits → corrective action. Framework: McKinsey 7S - Strategy, Structure, Systems, Shared values, Style, Staff, Skills
- Barriers: resistance to change, poor communication, lack of top support, weak resources, short-term thinking

Strategy evaluation - Balanced Scorecard (Kaplan & Norton)

- Financial (profit, ROI) - Customer (satisfaction, retention) - Internal processes (quality, cycle time) - Learning and growth (skills, innovation)
- Evaluate : Are assumptions valid? Is performance on target? Take corrective action (revise strategy, objectives or resources)

6. SWOT Analysis

SWOT (Albert Humphrey) = framework to match internal Strengths and Weaknesses with external Opportunities and Threats and choose strategy.

INTERNAL	EXTERNAL
STRENGTHS (+) brand, skilled staff, technology, cash, location	OPPORTUNITIES (+) growing market, new tech, policy support, export
WEAKNESSES (-) old machines, debt, weak R&D, high attrition	THREATS (-) competition, recession, new laws, tech change

TOWS strategy	With opportunities (O)	With threats (T)
Use strengths (S)	SO : use strengths to grab opportunities (growth)	ST : use strengths to defend threats (diversify)
Reduce weaknesses (W)	WO : overcome weaknesses with opportunities (turnaround)	WT : minimise weaknesses and avoid threats (retrench)

Using SWOT to strengthen a project

- 1. List S, W, O, T with evidence 2. Rank by importance 3. Match (TOWS) 4. Convert into actions with owners, dates 5. Review regularly
- Example (campus food-delivery start-up): S - low-cost student delivery force ; W - small capital ; O - 10,000 hostel students ; T - big apps with discounts → SO : tie-up with mess and canteens, student-only offers ; WT : stay in one campus first

7. Project Management

A project = temporary endeavour with a start and end, a unique output, specific objectives, limited resources and cost (e.g. building a bridge, launching an app). Project management = planning, organising, directing and controlling resources for a project to meet scope, time, cost and quality targets.

Project life cycle

- Conceptualisation or initiation → Planning (WBS, schedule, budget) → Execution → Monitoring and control → Closure or handover
- Tools : Gantt (bar) chart , milestone chart , network (PERT or CPM) , WBS, resource levelling. Triple constraint: scope = time = cost (quality at the centre)

Network analysis - basic terms

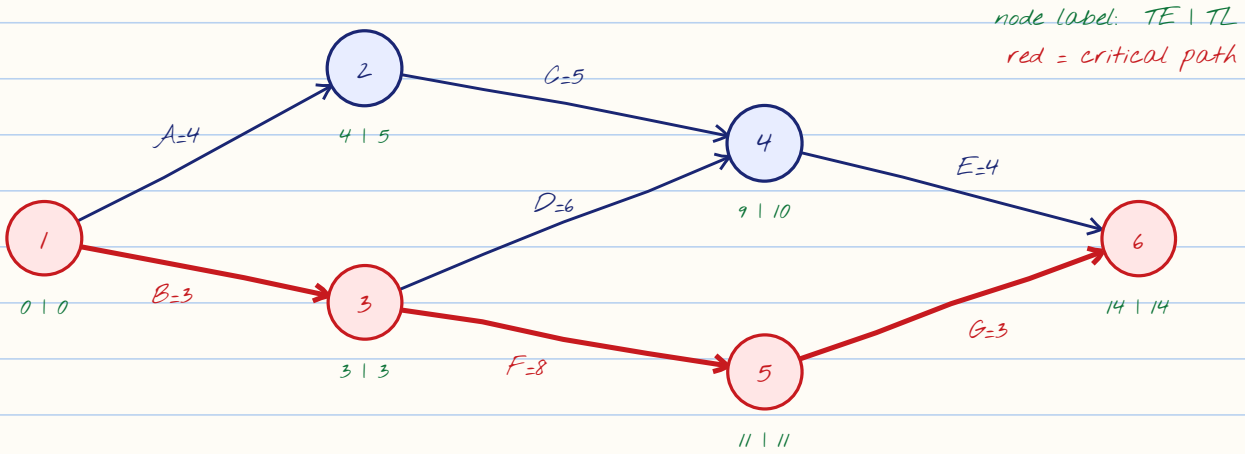
Term	Meaning
Activity	Time- and resource-consuming task, shown by an arrow (→)
Event (node)	Start or finish point of activities, shown by a circle ; consumes no time
Dummy activity	Dotted arrow, zero time - shows logical dependency only
Path	Continuous chain of activities from start to end event
Critical path	Longest path - decides project duration ; zero float
Float or slack	Spare time : Total float = LS - ES = LF - EF ; Free float, Independent float
Predecessor or successor	Activity that must precede or follow another

- Network rules : one start and one end event ; no loops ; events numbered in increasing order ($i < j$) ; no two activities share the same pair of events (use dummy)

Basis	CPM	PERT
Origin	DuPont and Remington Rand, 1957 (construction, chemical)	US Navy, Polaris missile, 1958 (R&D)
Time estimate	Single (deterministic)	Three (a, m, b) - probabilistic
Orientation	Activity-oriented	Event-oriented
Focus	Time-cost trade-off (crashing)	Time control, probability
Use	Repetitive, Known jobs	Non-repetitive, uncertain jobs

8. CPM - Identifying the Critical Path

Steps : draw network → forward pass (ES, EF = ES + t ; event TE = max) → backward pass (LF, LS = LF - t ; event TL = min) → float = LS - ES → critical path = activities with zero float.



Activity	Nodes	t	ES	EF	LS	LF	Float	Critical?
A	1-2	4	0	4	1	5	1	No
B	1-3	3	0	3	0	3	0	Yes
C	2-4	5	4	9	5	10	1	No
D	3-4	6	3	9	4	10	1	No
E	4-6	4	9	13	10	14	1	No
F	3-5	8	3	11	3	11	0	Yes
G	5-6	3	11	14	11	14	0	Yes

Result

- Paths : A-C-E = 13 ; B-D-E = 13 ; B-F-G = 3 + 8 + 3 = 14 days (longest)
  → Critical path = 1-3-5-6 (B-F-G), project duration = 14 days

Types of float (from the example)

Float	Formula	Meaning
Total float	LS - ES (= LF - EF)	Delay possible without delaying the project
Free float	TE(j) - TE(i) - t	Delay possible without affecting the next activity
Independent float	TE(j) - TL(i) - t	Delay possible even if predecessors are late

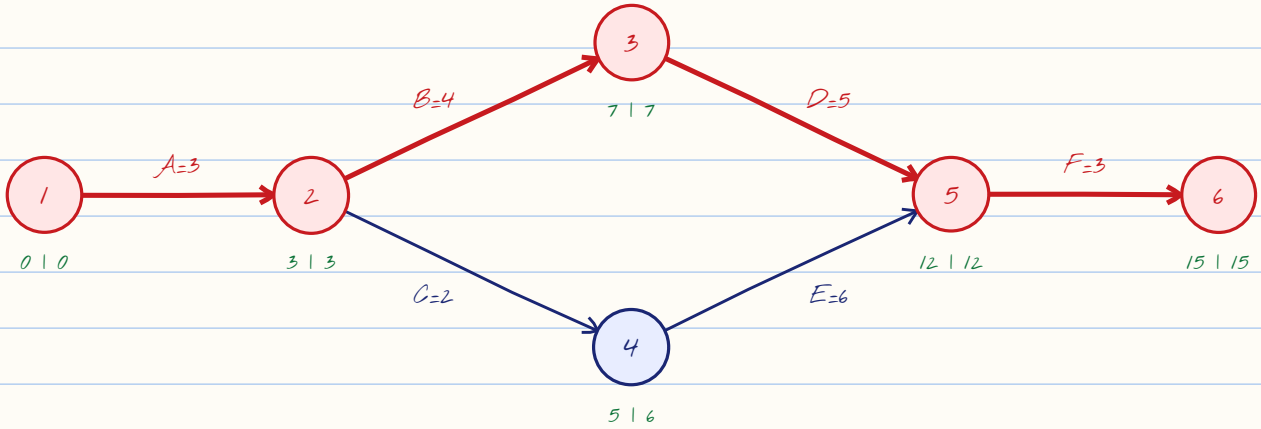
- Example : A → total float 1, free float 4 - 0 - 4 = 0 ; E → total float

1, free float = 14 - 9 - 4 = 1. Critical activities (B, F, G) have all floats = 0.

Second problem - from a precedence table

Activity	Predecessor	Time (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	6
F	D, E	3

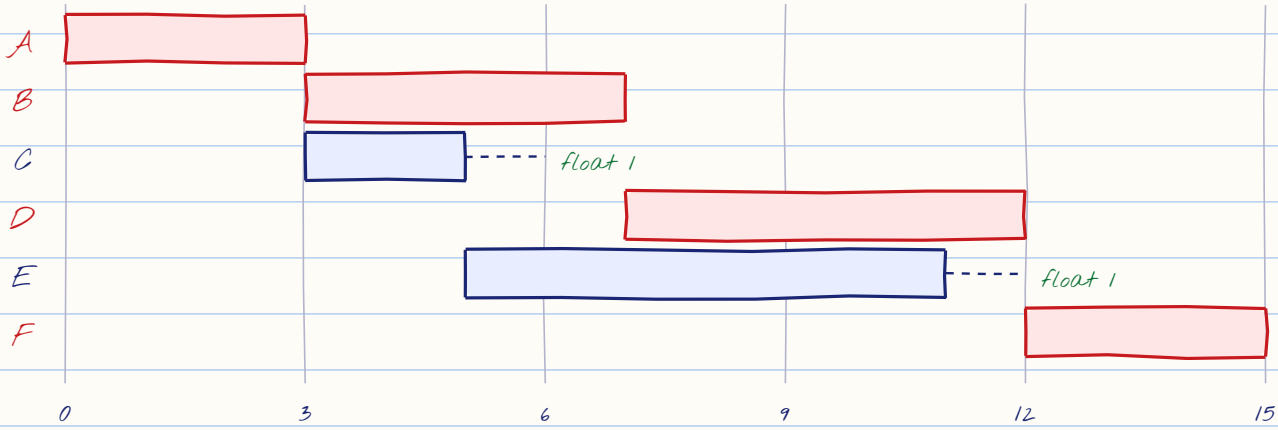
red = critical (A-B-D-F = 15)



Activity	t	ES	EF	LS	LF	Float
A	3	0	3	0	3	0
B	4	3	7	3	7	0
C	2	3	5	4	6	1
D	5	7	12	7	12	0
E	6	5	11	6	12	1
F	3	12	15	12	15	0

- Paths : A-B-D-F = 3 + 4 + 5 + 3 = 15 ; A-C-E-F = 3 + 2 + 6 + 3 = 14 → Critical path A-B-D-F, duration 15 days

Gantt (bar) chart of the same project



- Gantt chart shows when each activity starts and ends but not the dependency clearly ; network shows dependency, critical path and float. PERT or CPM merits : planning, control by exception, resource focus, what-if analysis ; demerits : needs time estimates, complex for large projects.

9. PERT - Probability of Completing in Given Time

Formulae

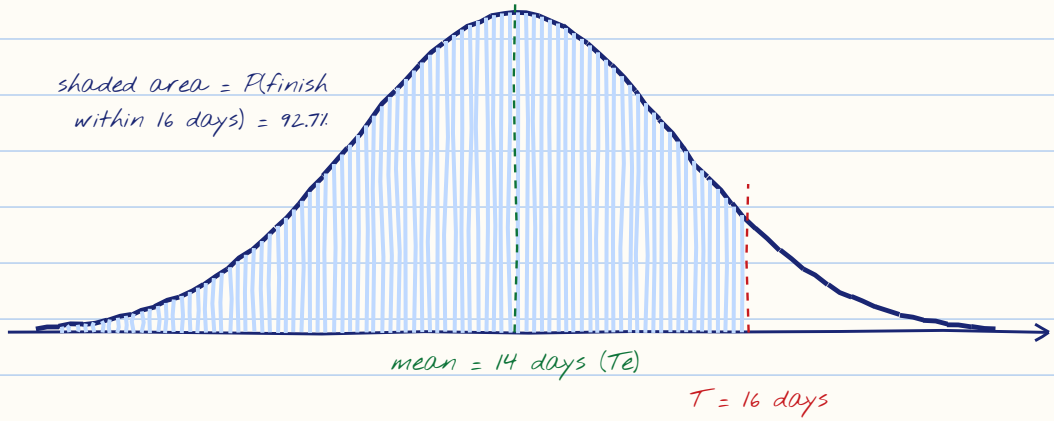
PERT formulae

- Expected time te = (a + 4m + b) ÷ 6 (a = optimistic, m = most likely, b = pessimistic)
- Variance = ((b − a) ÷ 6)² ; Std. deviation = sqrt(variance)
- Project mean Te = sum of te on critical path ; Project variance = sum of variances on critical path
- Z = (T − Te) ÷ sigma → probability from normal table (P = 0.5 when T = Te)

Activity	a	m	b	te	variance
B (1-3)	1	3	5	3	0.444
F (3-5)	5	8	11	8	1.000
G (5-6)	1	3	5	3	0.444
Total (critical)				Te = 14	1.889

Example - same network

- Project variance = 1.889 ; sigma = sqrt(1.889) = 1.374 days
- P(complete within 16 days) : Z = (16 - 14) ÷ 1.374 = 1.455 → P = 0.927 (92.7%)
- P(complete within 12 days) : Z = (12 - 14) ÷ 1.374 = -1.455 → P = 1 - 0.927 = 0.073 (7.3%)
- Duration for 95% confidence = 14 + 1.645 x 1.374 = 16.26 days



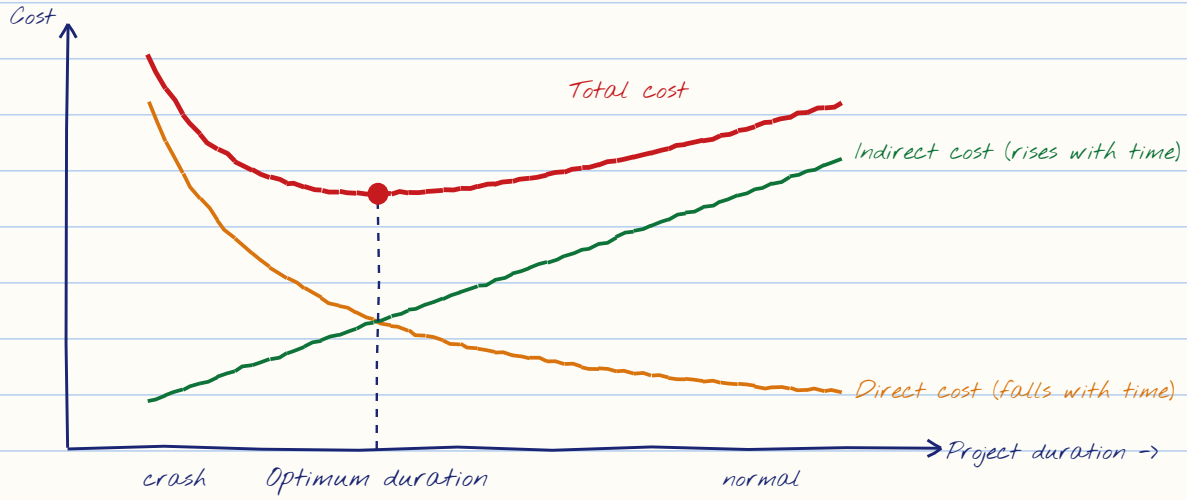
Second PERT problem (critical path A-B-D-F of the 15-day project)

Activity	a	m	b	te	variance
A	2	3	4	3	0.111
B	2	4	6	4	0.444
D	3	5	7	5	0.444
F	2	3	4	3	0.111
Total				15	1.111

- sigma = sqrt(1.111) = 1.054 ; P(finish in 17 days) : Z = (17 - 15) ÷ 1.054 = 1.90 → 0.971 (97.1%) ; P(finish in 14 days) : Z = -0.95 → 0.171 (17.1%)

10. Project Cost Analysis & Crashing

Crashing = shortening project time by adding resources to critical activities at least extra cost. Direct cost rises when time is cut ; indirect cost (overheads, supervision, penalty) falls.



Terms & formula

- Normal time or cost (Tn, Cn) ; Crash time or cost (Tc, Cc)
- Cost slope = (Cc − Cn) ÷ (Tn − Tc) = extra cost per day saved
- Steps : find critical path → crash the critical activity with lowest cost slope (by 1 day each time) → recheck for new critical paths → stop when slope > indirect cost saved per day, or when no activity can be crashed

Activity	Normal (days)	Crash (days)	Normal cost (Rs)	Crash cost (Rs)	Cost slope (Rs/day)
A (1–2)	4	3	400	500	100
B (1–3)	5	3	500	700	100
C (2–4)	5	3	600	900	150
D (3–4)	3	2	300	420	120
E (4–5)	5	4	500	650	150

Network: A(1–2), B(1–3), C(2–4), D(3–4), E(4–5). Paths : A–C–E = 14 days (critical) and B–D–E = 13 days. Direct cost (normal) = Rs.2300. Indirect cost = Rs.130 per day.

Step	Action (crash)	Days	Direct (Rs)	Indirect (Rs)	Total (Rs)
0	Normal schedule	14	2300	1820	4120
1	A by 1 day (slope 100)	13	2400	1690	4090
2	E by 1 day (slope 150, common)	12	2550	1560	4110

Step	Action (crash)	Days	Direct (Rs)	Indirect (Rs)	Total (Rs)
3	B and C by 1 day each (100 + 150)	11	2800	1430	4230

Conclusion

- Total cost is lowest (Rs.4090) at 13 days - beyond this, crashing costs more than the indirect saving (Rs.130/day). Optimum duration = 13 days , optimum cost = Rs.4090 .

11. Important Questions (Unit IV)

1. Define strategy. Explain vision, mission, goals and objectives with examples.
2. Explain the corporate planning process and environmental scanning.
3. Explain the steps in strategy formulation and implementation.
4. What is SWOT analysis? Explain with an example.
5. Differentiate PERT and CPM. Explain network terms.
6. Solve a CPM problem : find ES, EF, LS, LF, float and critical path.
7. PERT problem : find expected time, variance, and probability of completion in given time.
8. Explain project crashing with a simple problem.

Last-minute revision

- te = (a + 4m + b) ÷ 6 ; variance = ((b − a) ÷ 6)² ; Z = (T − Te) ÷ sigma ; float = LS − ES ; critical path = longest path (zero float) ; cost slope = (Cc − Cn) ÷ (Tn − Tc)